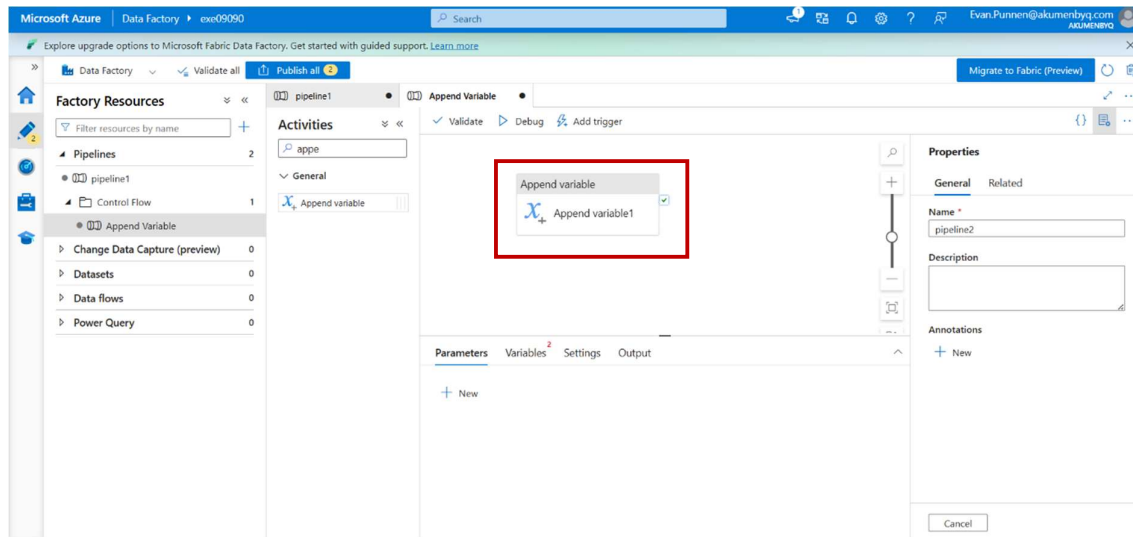
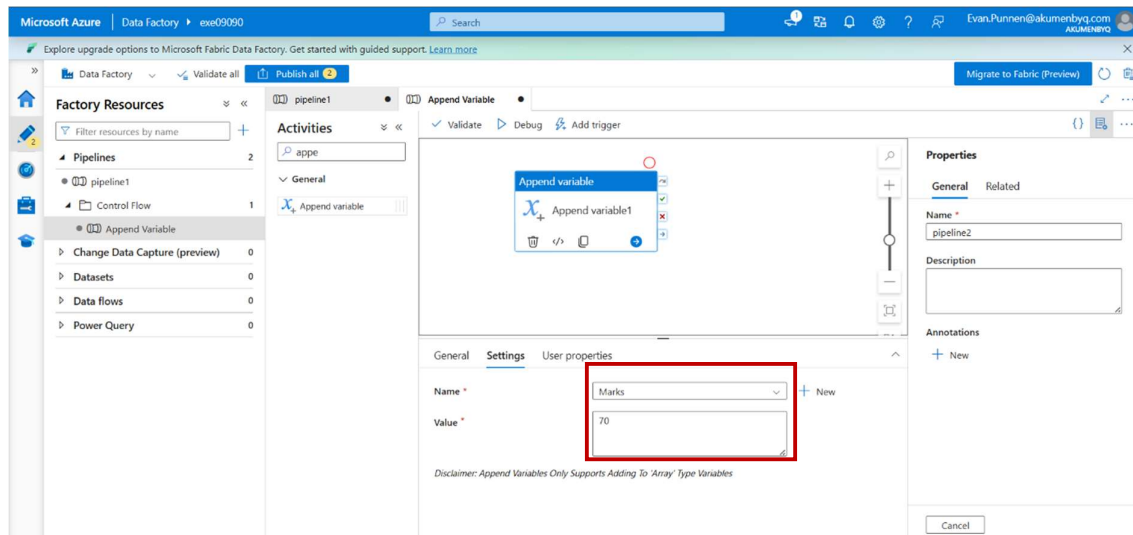


MICROSOFT AZURE

- Add the Append Variable activity to the pipeline



- Set the Marks variable and append the value 70



MICROSOFT AZURE

- Create an Array variable named Marks with default values.

The screenshot shows the Microsoft Azure Data Factory interface. In the 'Activities' pane, the 'Append Variable' activity is selected. The 'Variables' tab is active, displaying a table with the following data:

Name	Type	Default value
Marks	Array	10,20,30
Array	Array	Value

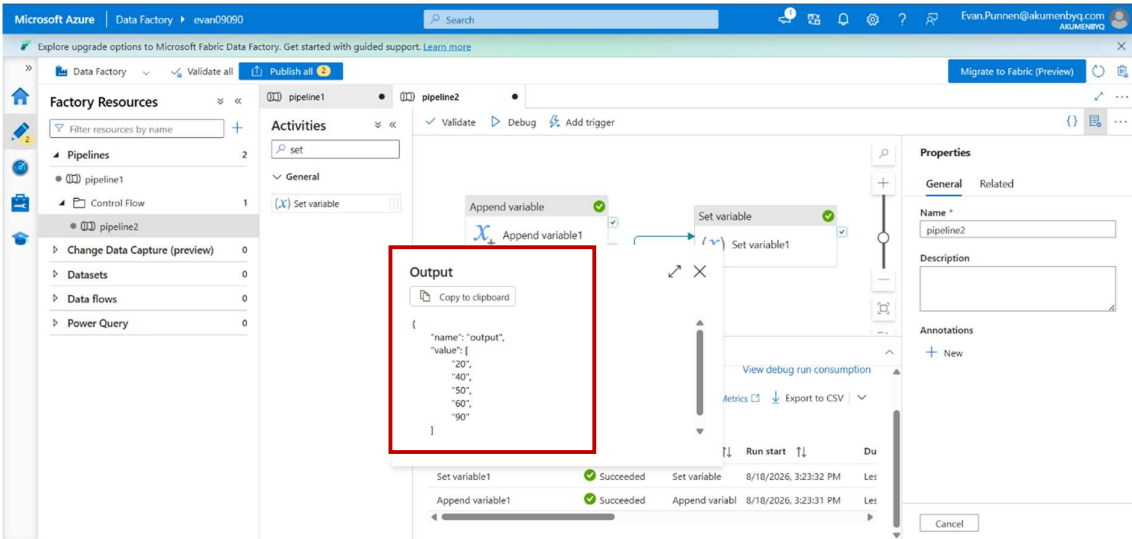
- Run the Append Variable pipeline and verify successful execution.

The screenshot shows the Microsoft Azure Data Factory interface. In the 'Activities' pane, the 'Append Variable' activity is selected. The 'Output' tab is active, displaying a table with the following data:

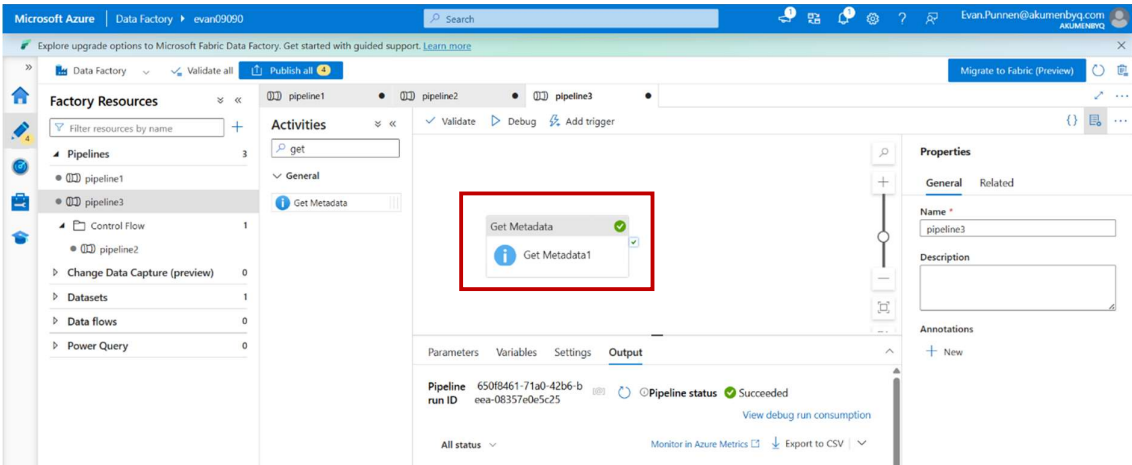
Activity name	Activity status	Activity name	Run start	Duration
Set variable1	Succeeded	Set variable	8/18/2026, 3:23:32 PM	Les
Append variable1	Succeeded	Append variabl	8/18/2026, 3:23:31 PM	Les

MICROSOFT AZURE

- Verify the appended Marks values in the pipeline output.

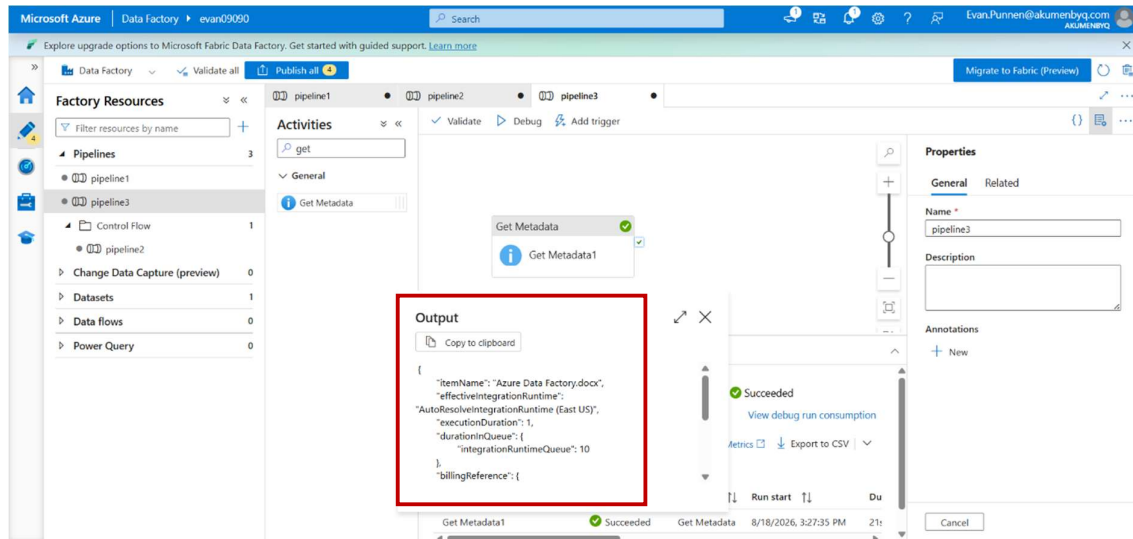


- Add and execute the Get Metadata activity.

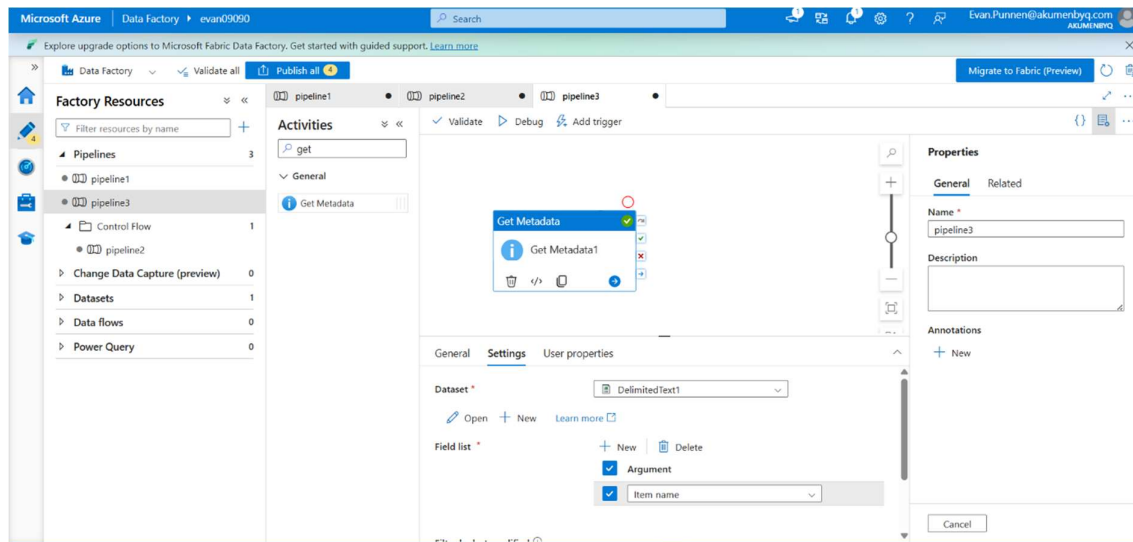


MICROSOFT AZURE

- Verify the metadata output for the source file.



- Configure Get Metadata with the Delimited Text dataset and Item Name field.



MICROSOFT AZURE

- **Connect Get Metadata to Set Variable and verify successful execution.**

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines: pipeline1, pipeline2, and pipeline3. The 'Activities' pane for pipeline1 shows a sequence of 'Get Metadata' and 'Set variable' activities. The main canvas shows the pipeline design with these two activities connected. The 'Output' tab at the bottom shows the execution results for the 'Set variable' and 'Get Metadata' activities, both marked as 'Succeeded'. On the right, a 'Pipeline validation output' window is open, displaying a message: 'Your pipeline has been validated. No errors were found.'

- **Add Execute Pipeline activity and select the child pipeline.**

The screenshot shows the Microsoft Azure Data Factory interface with the 'Execute Pipeline' activity added to the pipeline. The 'Activities' pane on the left shows the 'Execute Pipeline' activity selected. The main canvas shows the pipeline design with the 'Execute Pipeline' activity added. The 'Output' tab at the bottom shows the execution results for the 'Execute Pipeline' activity, indicating it was 'Succeeded'. The 'Properties' pane on the right shows the configuration for the 'Execute Pipeline' activity, including the 'Name' field set to 'pipeline1' and the 'Description' field.

MICROSOFT AZURE

- Validate the pipeline and verify the metadata Count output.

The screenshot displays the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane shows a list of pipelines: pipeline1, pipeline2, and pipeline3. The 'Activities' pane for pipeline1 shows a 'Set variable' activity. The main canvas shows a pipeline diagram with a 'Get Metadata' activity. A red box highlights the 'Output' of the 'Get Metadata' activity, which is a JSON object:

```
{ "name": "Count", "value": 3 }
```

. On the right, a 'Pipeline validation output' pane shows a message: 'Your pipeline has been validated. No errors were found.'

- Configure Execute Pipeline to run pipeline3.

The screenshot displays the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane shows a list of pipelines: pipeline1, pipeline2, and pipeline3. The 'Activities' pane for pipeline1 shows an 'Execute Pipeline' activity. The main canvas shows a pipeline diagram with an 'Execute Pipeline' activity. The 'Properties' pane on the right shows the configuration for the 'Execute Pipeline' activity, including the 'Name' (pipeline1) and 'Description'. The 'Output' tab shows the 'Pipeline status' as 'Succeeded'.

MICROSOFT AZURE

- Run the Execute Pipeline activity and verify successful execution.

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines, including 'pipeline1', 'pipeline3', 'pipeline4', and 'Control Flow'. The 'Activities' pane for 'pipeline1' shows a 'Set variable' activity. The main canvas shows a workflow with an 'Execute Pipeline' activity (labeled 'Execute Pipeline1 pipeline4') followed by a 'Set variable' activity (labeled 'Set variable1'). The 'Output' tab is active, showing a table with the following data:

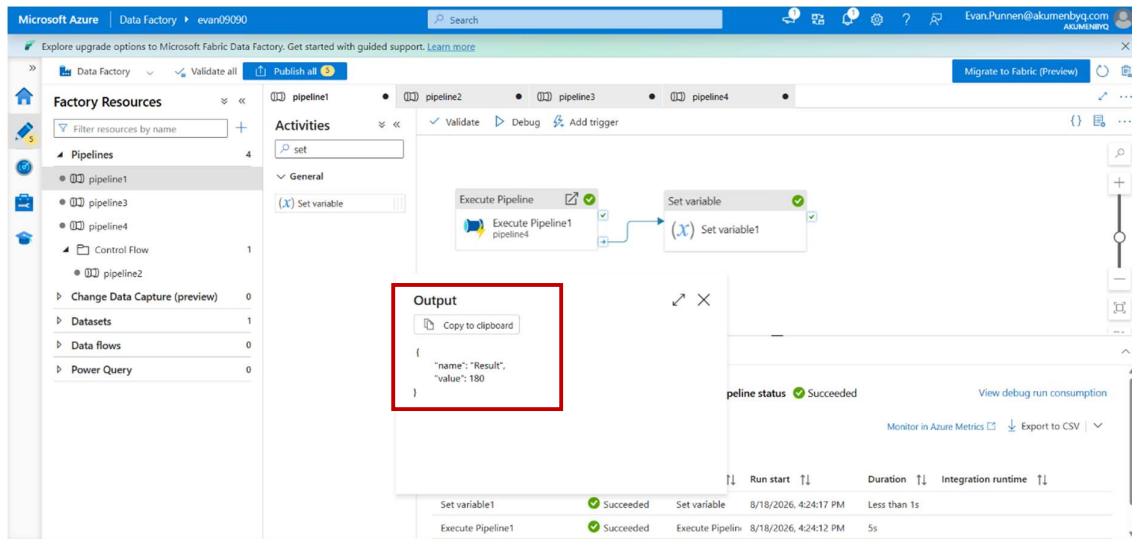
Activity name	Activity status	Activity name	Run start	Duration
Set variable1	Succeeded	Set variable	8/18/2026, 4:24:17 PM	Less
Execute Pipeline1	Succeeded	Execute Pipeline1	8/18/2026, 4:24:12 PM	5s

- Configure Set Variable to store the calculated Result value.

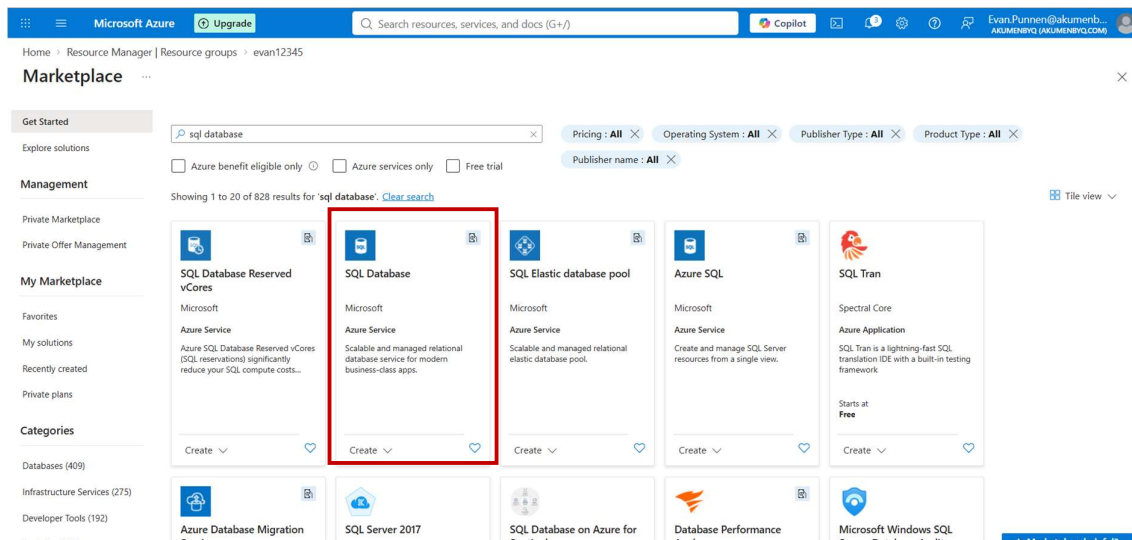
The screenshot displays the Microsoft Azure Data Factory interface, focusing on the configuration of a 'Set variable' activity. The 'Settings' tab is active, showing the 'Variable type' set to 'Pipeline variable'. The 'Name' field is set to 'Result', and the 'Value' field is set to '@mul(pipeline().parameters.Length, pipeline().parameters.Length)'. The 'Output' tab is also visible, showing the same data as the previous screenshot.

MICROSOFT AZURE

- Verify the calculated Result value as 180.



- Search for SQL Database in the Azure Marketplace.



MICROSOFT AZURE

- Configure the SQL Server name, location, and authentication method.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Resource Manager | Resource groups > evan12345 > Marketplace > SQL Database > Create SQL Database

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * .database.windows.net

Location *

Authentication

! Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) if using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#) if

Authentication method

☒ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☐ Use SQL authentication

OK Feedback

- Review the SQL Database configuration and cost details.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Resource Manager | Resource groups > evan12345 > Marketplace > SQL Database

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Product details

SQL database by Microsoft [Terms of use](#) | [Privacy policy](#)

Estimated cost per month

Compute cost -- USD + Storage cost 0.30 USD / GB

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription Azure subscription 1

Resource group evan12345

Region India South Central

Database name data989898

Server (new) serverdb00

Create < Previous Download a template for automation

Cost summary

Hyperscale (HS_Gen5_2) - Primary replica

Cost per vCore (in --)

vCores selected x 2

Hyperscale (HS_Gen5_2) - HA replicas

Cost per vCore (in --)

vCores selected x 2

HA replicas x 0

ESTIMATED COMPUTE COST / MONTH -- USD

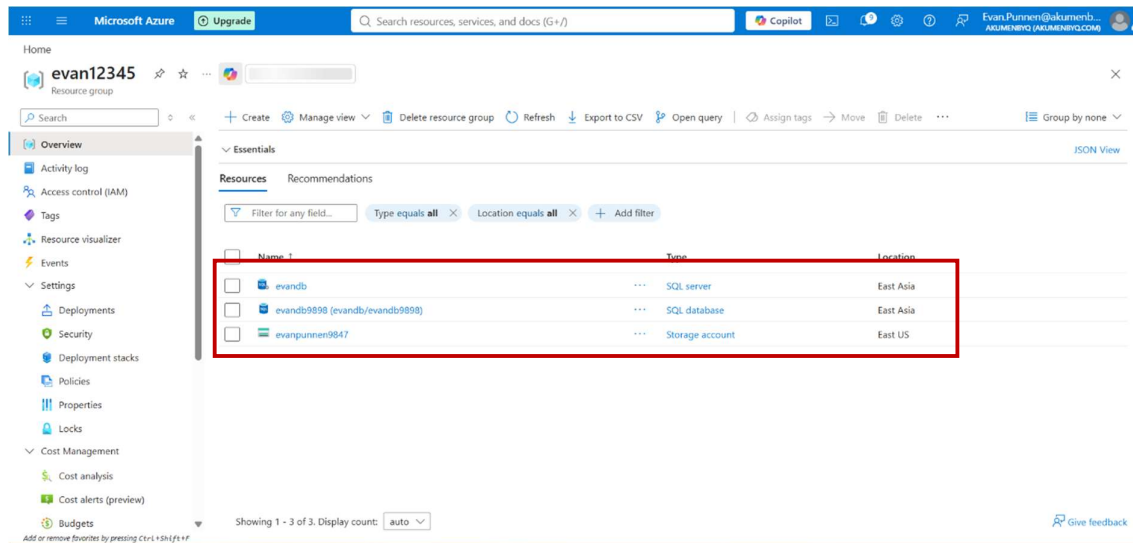
STORAGE COST / GB / MONTH 0.30 USD

STORAGE NOTES

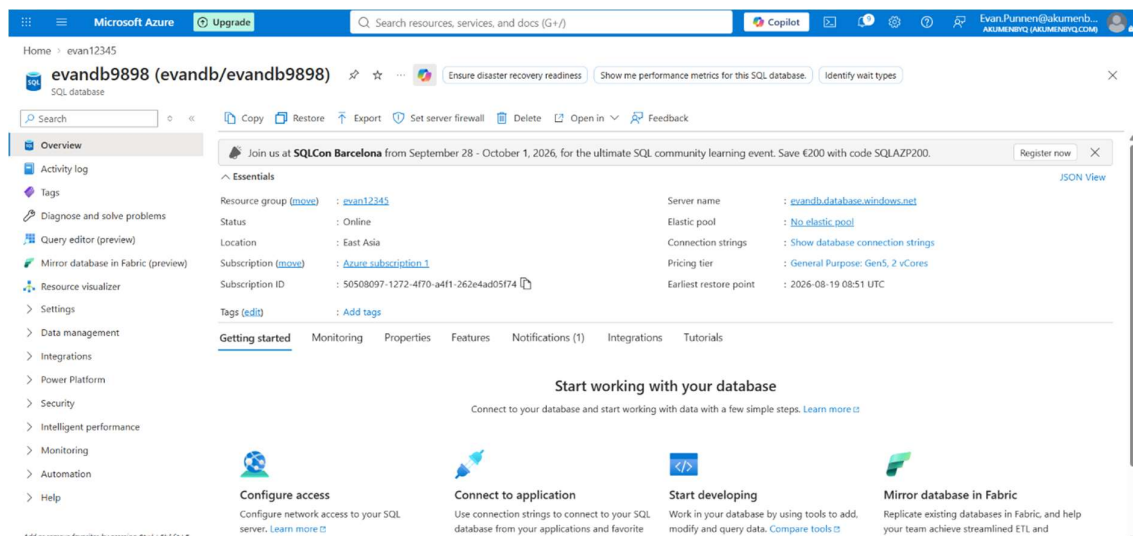
¹ In the Hyperscale tier, storage costs are calculated based on actual allocation. Allocated space increases

MICROSOFT AZURE

- Verify the created SQL Server, SQL Database, and Storage Account resources.

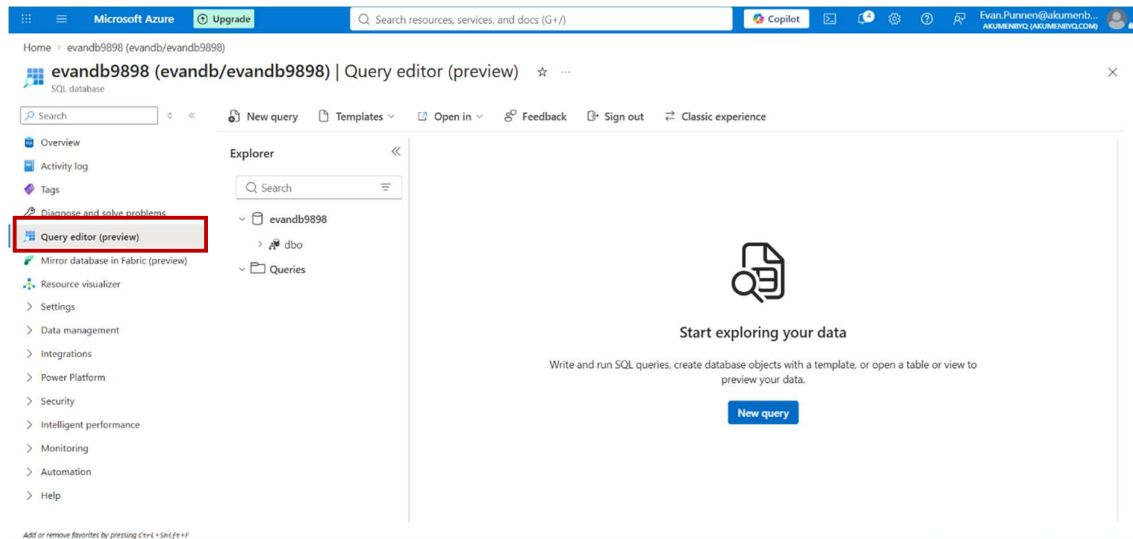


- Open the created Azure SQL Database overview.

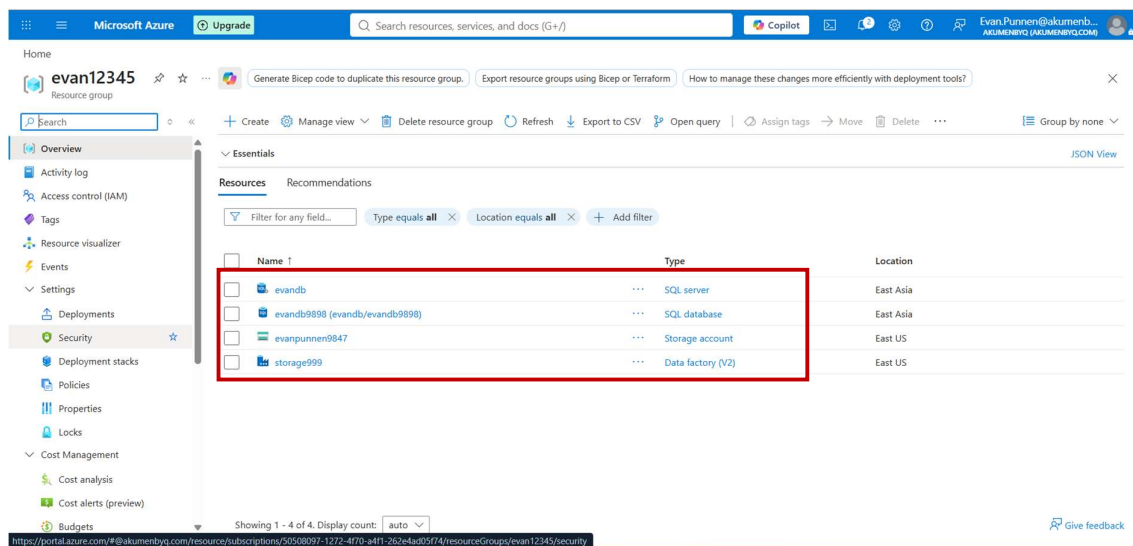


MICROSOFT AZURE

- Open the SQL Database Query Editor.

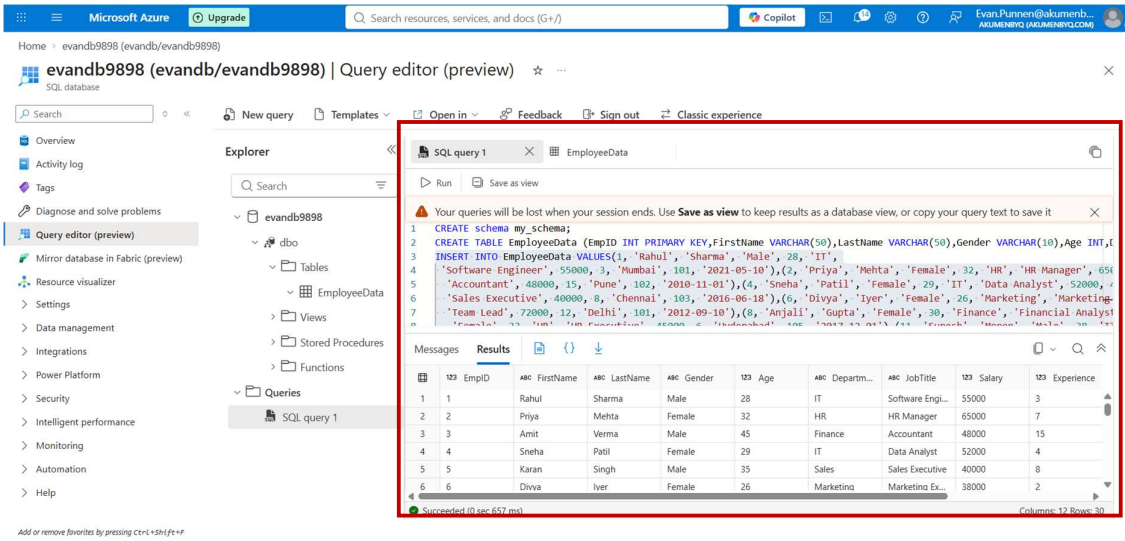


- Open the resource group and verify the Azure resources.



MICROSOFT AZURE

- Create the Employee Data table, insert records, and execute the SQL query.

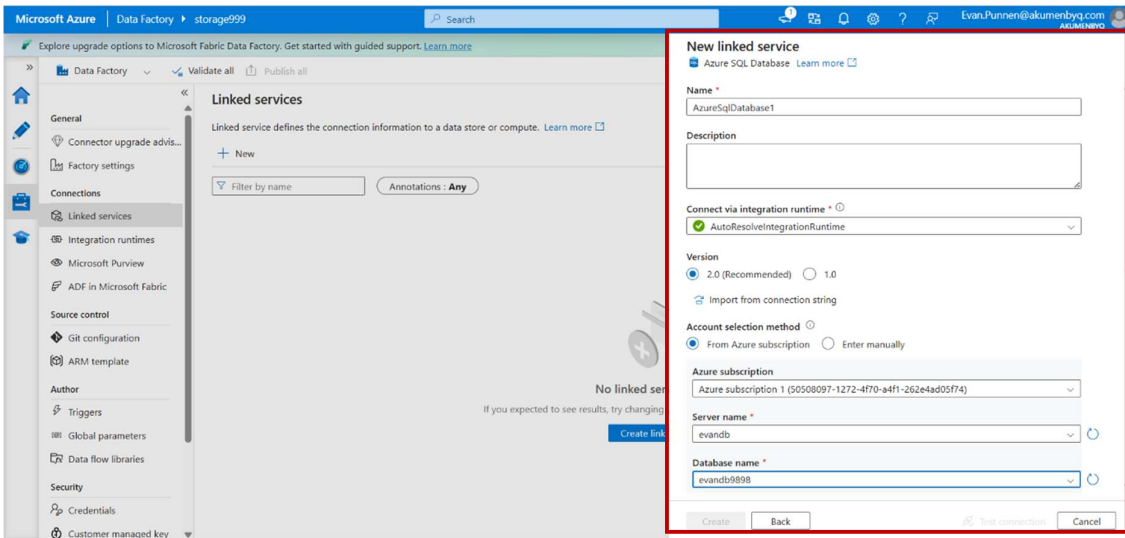


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a user profile. The main content area is titled 'evandb9898 (evandb/evandb9898) | Query editor (preview)'. On the left, there is a sidebar with navigation options like Overview, Activity log, Tags, and Query editor (preview). The central pane shows the SQL query editor with a query that creates a table 'EmployeeData' and inserts 6 records. The results are displayed in a table below the query.

```
1 CREATE schema my_schema;
2 CREATE TABLE EmployeeData (EmpID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50), Gender VARCHAR(10), Age INT, IT
3 INSERT INTO EmployeeData VALUES(1, 'Rahul', 'Sharma', 'Male', 28, 'IT',
4 ('Software Engineer', 55000, 3, 'Mumbai', 101, '2021-05-10'), (2, 'Priya', 'Mehta', 'Female', 32, 'HR', 'HR Manager', 65000, 3,
5 'Accountant', 48000, 15, 'Pune', 102, '2010-11-01'), (4, 'Sneha', 'Patil', 'Female', 29, 'IT', 'Data Analyst', 52000, 4,
6 'Sales Executive', 40000, 8, 'Chennai', 103, '2016-06-18'), (6, 'Divya', 'Iyer', 'Female', 26, 'Marketing', 'Marketing Ex...', 38000, 2,
7 'Team Lead', 72000, 12, 'Delhi', 101, '2012-09-10'), (8, 'Anjali', 'Gupta', 'Female', 30, 'Finance', 'Financial Analyst', 60000, 1,
8 'Team Lead', 72000, 12, 'Delhi', 101, '2012-09-10'), (8, 'Anjali', 'Gupta', 'Female', 30, 'Finance', 'Financial Analyst', 60000, 1,
```

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience
1	Rahul	Sharma	Male	28	IT	Software Engineer	55000	3
2	Priya	Mehta	Female	32	HR	HR Manager	65000	7
3	Amit	Verma	Male	45	Finance	Accountant	48000	15
4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4
5	Karan	Singh	Male	35	Sales	Sales Executive	40000	8
6	Divya	Iyer	Female	26	Marketing	Marketing Executive	38000	2

- Create an Azure SQL Database linked service in Data Factory.



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area is titled 'Data Factory | storage999'. On the left, there is a sidebar with navigation options like Data Factory, General, Connector upgrade advisor, Factory settings, Connections, Linked services, Integration runtimes, Microsoft Purview, ADF in Microsoft Fabric, Source control, Git configuration, ARM template, Author, Triggers, Global parameters, Data flow libraries, Security, Credentials, and Customer managed key. The central pane shows the 'Linked services' section. A 'New linked service' dialog is open, showing the configuration for an Azure SQL Database linked service.

New linked service

Azure SQL Database [Learn more](#)

Name *
AzureSqlDatabase1

Description

Connect via integration runtime *
☒ AutoResolveIntegrationRuntime

Version
☒ 2.0 (Recommended) ☐ 1.0

Import from connection string

Account selection method
☒ From Azure subscription ☐ Enter manually

Azure subscription
Azure subscription 1 (50508097-1272-4f70-a4f1-262e4ad05f74)

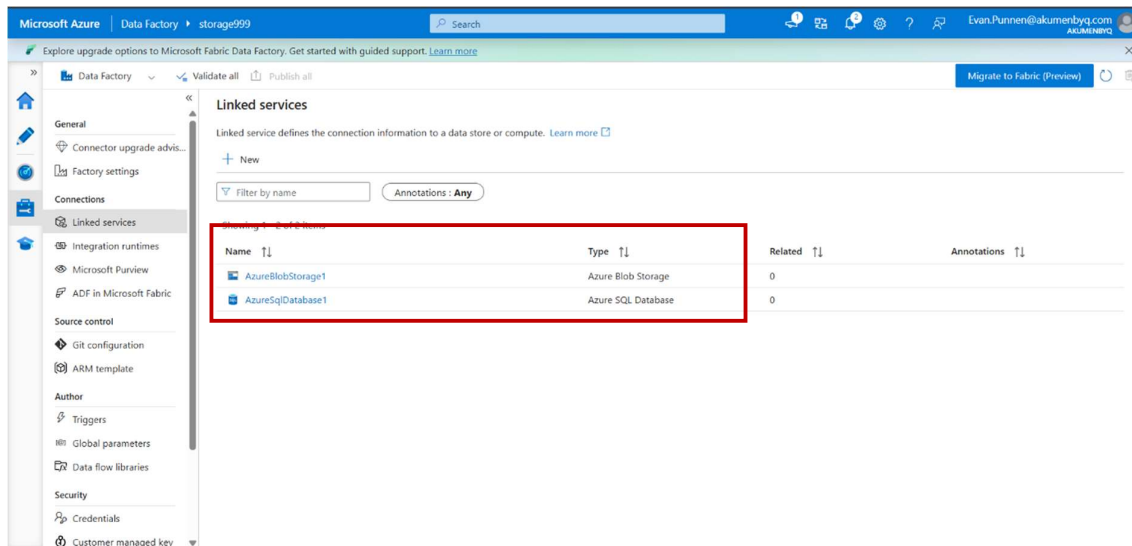
Server name *
evandb

Database name *
evandb9898

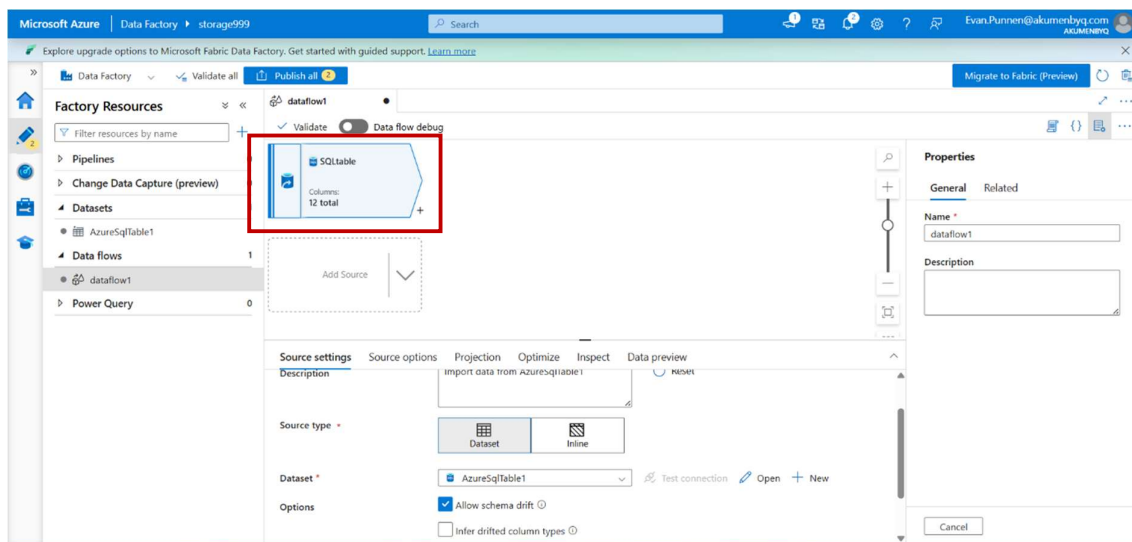
Create Back [For subscription](#) Cancel

MICROSOFT AZURE

- Verify the created Azure SQL Database and other linked services.

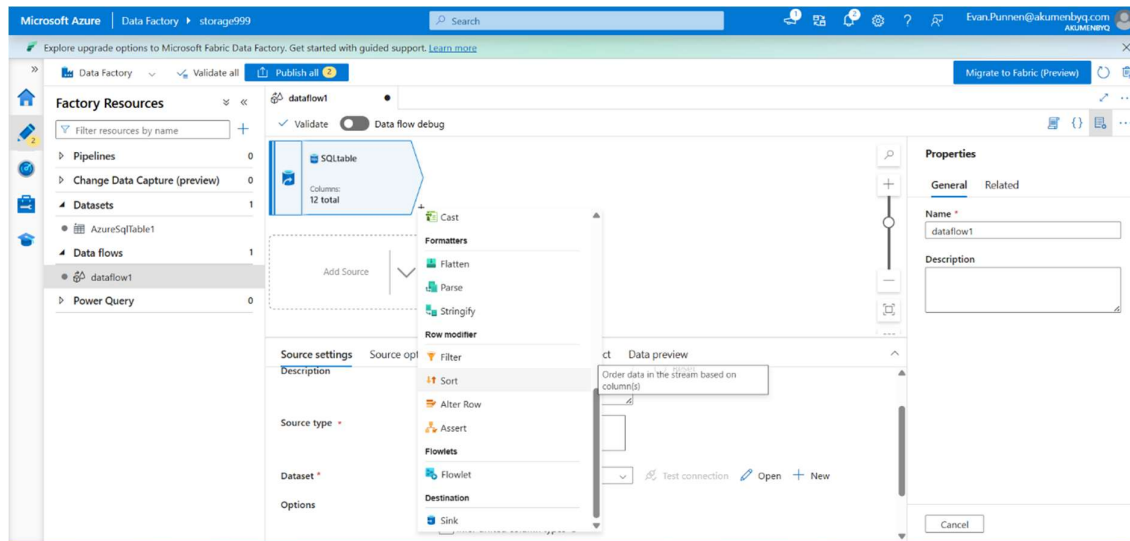


- Configure the Data Flow source using the AzureSqlTable1 dataset.

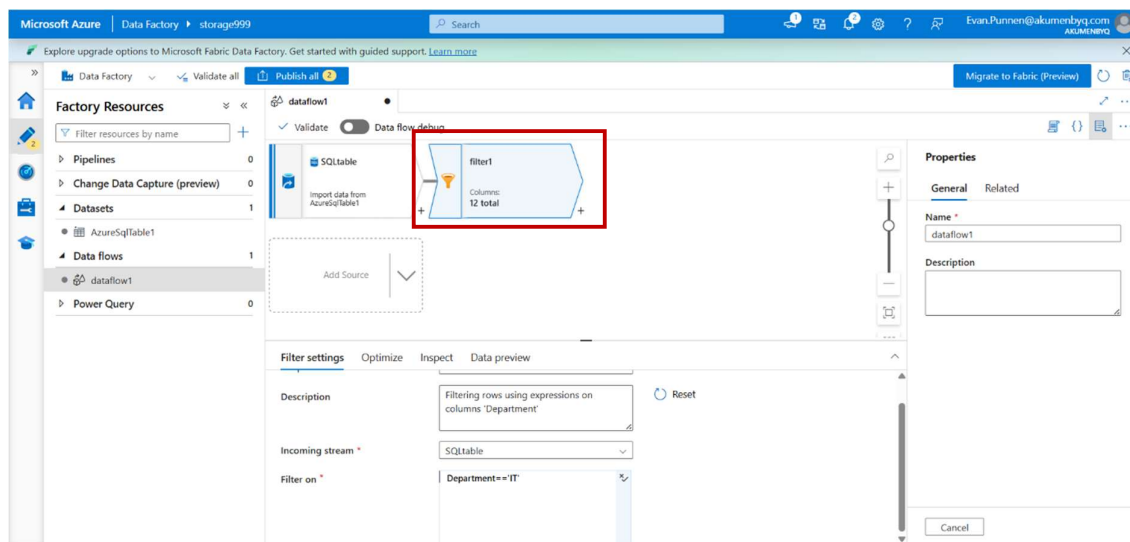


MICROSOFT AZURE

- Open the transformation menu and select the Filter transformation.



- Configure the Filter condition to select Department = IT.



MICROSOFT AZURE

- Enable Data Flow Debug and preview the filtered records.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Data flows' with 'dataflow1' selected. The main workspace shows a data flow diagram with a source 'sqltable' and a filter 'filter1'. The 'Data preview' tab is active, displaying a table with 9 rows. The first three rows are highlighted with a red box, indicating the filtered results.

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle
1	Rahul	Sharma	Male	28	IT	Software En
4	Sneha	Patil	Female	29	IT	Data Analys
7	Rohit	Nair	Male	41	IT	Team Lead

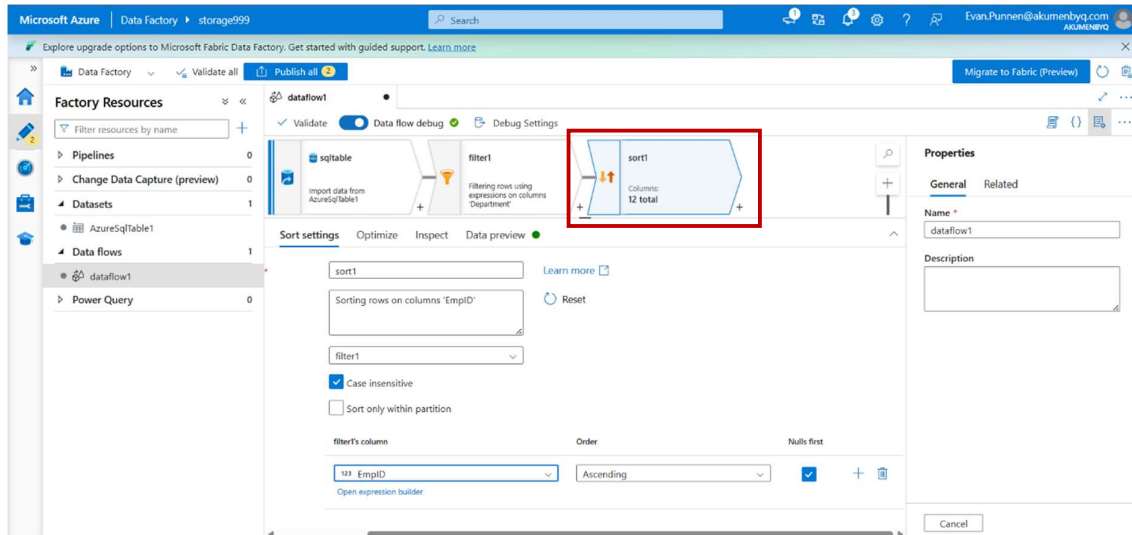
- Verify that the preview contains only IT department records.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Data flows' with 'dataflow1' selected. The main workspace shows a data flow diagram with a source 'sqltable' and a filter 'filter1'. The 'Data preview' tab is active, displaying a table with 9 rows. The first three rows are highlighted with a red box, indicating the filtered results.

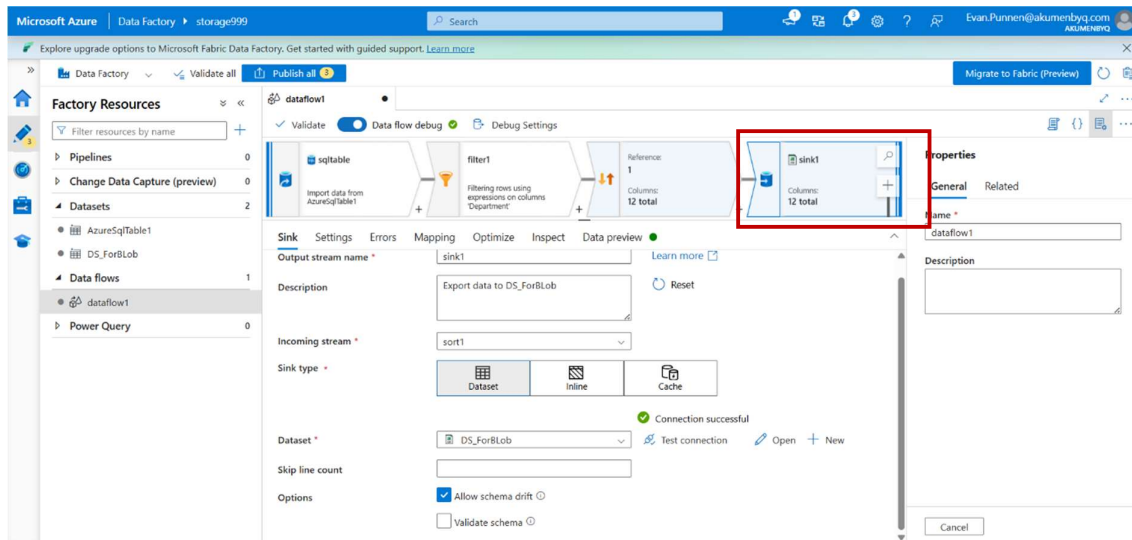
EmpID	FirstName	LastName	Gender	Age	Department	JobTitle
1	Rahul	Sharma	Male	28	IT	Software En
4	Sneha	Patil	Female	29	IT	Data Analys
7	Rohit	Nair	Male	41	IT	Team Lead

MICROSOFT AZURE

- Add Sort transformation and sort EmpID in ascending order.

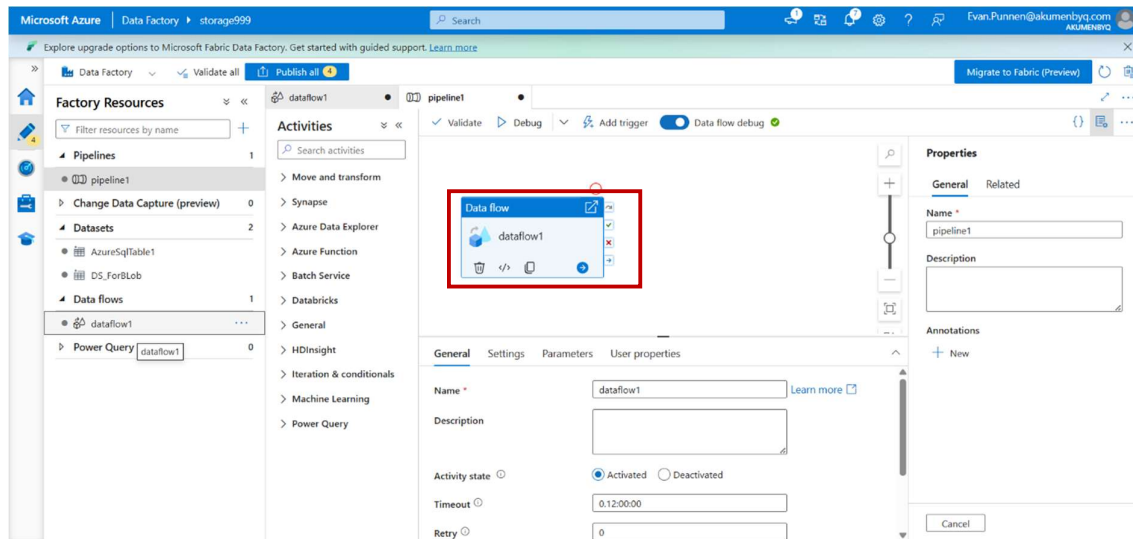


- Create a new dataset for the Data Flow sink.
- Configure the sink using the DS_ForBlob dataset.

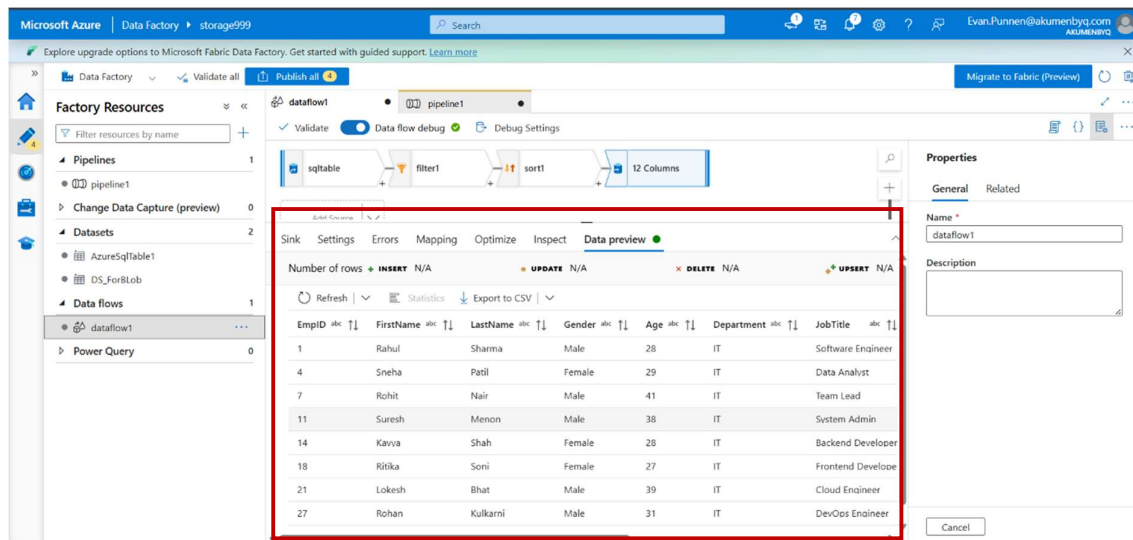


MICROSOFT AZURE

- Add the Data Flow activity to the pipeline and select dataflow1.



- Preview the final filtered and sorted Data Flow output.



MICROSOFT AZURE

- Open the output container and verify the generated files.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > evan12345 > evanpunner9847 | Containers

output

Container

Search

+ Add Directory | Upload | Change access level | Refresh | Delete | Copy | Paste | Rename | Acquire lease | Break lease | Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 3 items

	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	_SUCCESS	8/19/2026, 3:22:22 PM	Hot (Inferred)	Block blob	0	Available	...
☒	part-00000-011a9fe9-1b86-4aa0-952b-688af5d73bcc-c000.csv	8/19/2026, 3:20:20 PM	Hot (Inferred)	Block blob	759 B	Available	...
☐	part-00000-c391c19e-14f5-4e31-aeaa-ff3f9c8a03fa-c000.csv	8/19/2026, 3:22:22 PM	Hot (Inferred)	Block blob	759 B	Available	...

Add or remove favorites by pressing Ctrl+Shift+F

- Open the generated output file and verify the exported data

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > evan12345 > evanpunner9847 | Containers

output

Container

Search

+ Add Directory

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by

Showing all 3 items

Name

_SUCCESS

part-00000-011a9fe9-1b86-4aa0-952b-688af5d73bcc-c000.csv

part-00000-c391c19e-14f5-4e31-aeaa-ff3f9c8a03fa-c000.csv

part-00000-011a9fe9-1b86-4aa0-952b-688af5d73bcc-c000.csv

Save | Discard | Download | Refresh | Delete

Overview | Versions | Snapshots | Edit | Generate SAS

```
1 EmpID,Firstname,Lastname,Gender,Age,Department,JobTitle,Salary,Experience,City,ManagerID,JoinDate
2 1,Rahul,Sharma,Male,28,IT,Software Engineer,55000,3,Mumbai,101,2021-05-10
3 4,Sneha,Patil,Female,29,IT,Data Analyst,52000,4,Bangalore,101,2020-08-20
4 7,Rohit,Nair,Male,41,IT,Team Lead,72000,12,Delhi,101,2012-09-10
5 11,Suresh,Menon,Male,38,IT,System Admin,50000,10,Bangalore,101,2014-07-19
6 14,Kavya,Shah,Female,28,IT,Backend Developer,56000,4,Pune,101,2019-06-15
7 18,Ritika,Soni,Female,27,IT,Frontend Developer,54000,3,Bangalore,101,2021-03-09
8 21,Lokesh,Bhat,Male,39,IT,Cloud Engineer,68000,11,Hyderabad,101,2014-02-17
9 27,Rohan,Kulkarni,Male,31,IT,DevOps Engineer,61000,6,Bangalore,101,2017-05-29
10 30,Reema,Jain,Female,29,IT,QA Engineer,50000,5,Chennai,101,2019-12-19
11
```

Csv | Preview

Add or remove favorites by pressing Ctrl+Shift+F

MICROSOFT AZURE

- Open the generated output file and verify the exported data

The screenshot shows the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane lists 'Pipelines' with 'pipeline1' selected. The main area displays the 'Data flow' activity 'dataflow1' within 'pipeline1'. The 'Output' tab shows the pipeline status as 'Succeeded' with a run ID of 'a888c062-a792-4a9f-836d-ecfcd237625e'. Below this, a table shows the activity details:

Activity name	Activity st...	Activit...	Run start	Durati...
dataflow1	Succeeded	Data flow	8/19/2026, 3:21:40 PM	1m 3s

- Create a second Data Flow using two SQL sources, an Inner Join, and a sink, then preview the joined data.

The screenshot shows the Microsoft Azure Data Factory console with a new data flow 'dataflow2' being created. The visual designer shows two sources ('source1' and 'source2') connected to an inner join activity ('join1'), which is then connected to a sink activity ('sink1'). The 'Data preview' tab is active, showing a table with 2 rows of data:

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle
20	Meera	Pai	Female	25	Sales	Sales Trainee
27	Rohan	Kulkarni	Male	31	IT	DevOps Engineer